

Example 11 of Day-Street "Monoidal bicategories..."

We will take the antipode axiom at the bottom of page 142 in the form of a $\mathcal{V}$ -functor $S: \mathcal{H} \rightarrow \mathcal{H}^{\text{op}}$ together with a $\mathcal{V}$ -natural isomorphism

$$(*) \quad \mathcal{H}(B, C) \otimes \mathcal{H}(A, SB) \cong \mathcal{H}(B, C) \otimes \mathcal{H}(A, C).$$

Proposition Each Hopf monoid H in a braided monoidal category $\mathcal{V}$ gives a one-object example of a Hopf $\mathcal{V}$ -algebroid.

Proof Here $\mathcal{H}$ has one object A and $\mathcal{H}(A, A) = H$ so that $(*)$ becomes an isomorphism

$$H \otimes H \cong H \otimes H$$

for which we take the fusion map N . To see that it gives a $(*)$ which is $\mathcal{V}$ -natural, we must see what functoriality of $(*)$ in A, B, C means on the two sides. Both sides become left H -, right $H \otimes H$ -modules. The left $H \otimes H$ has action given by



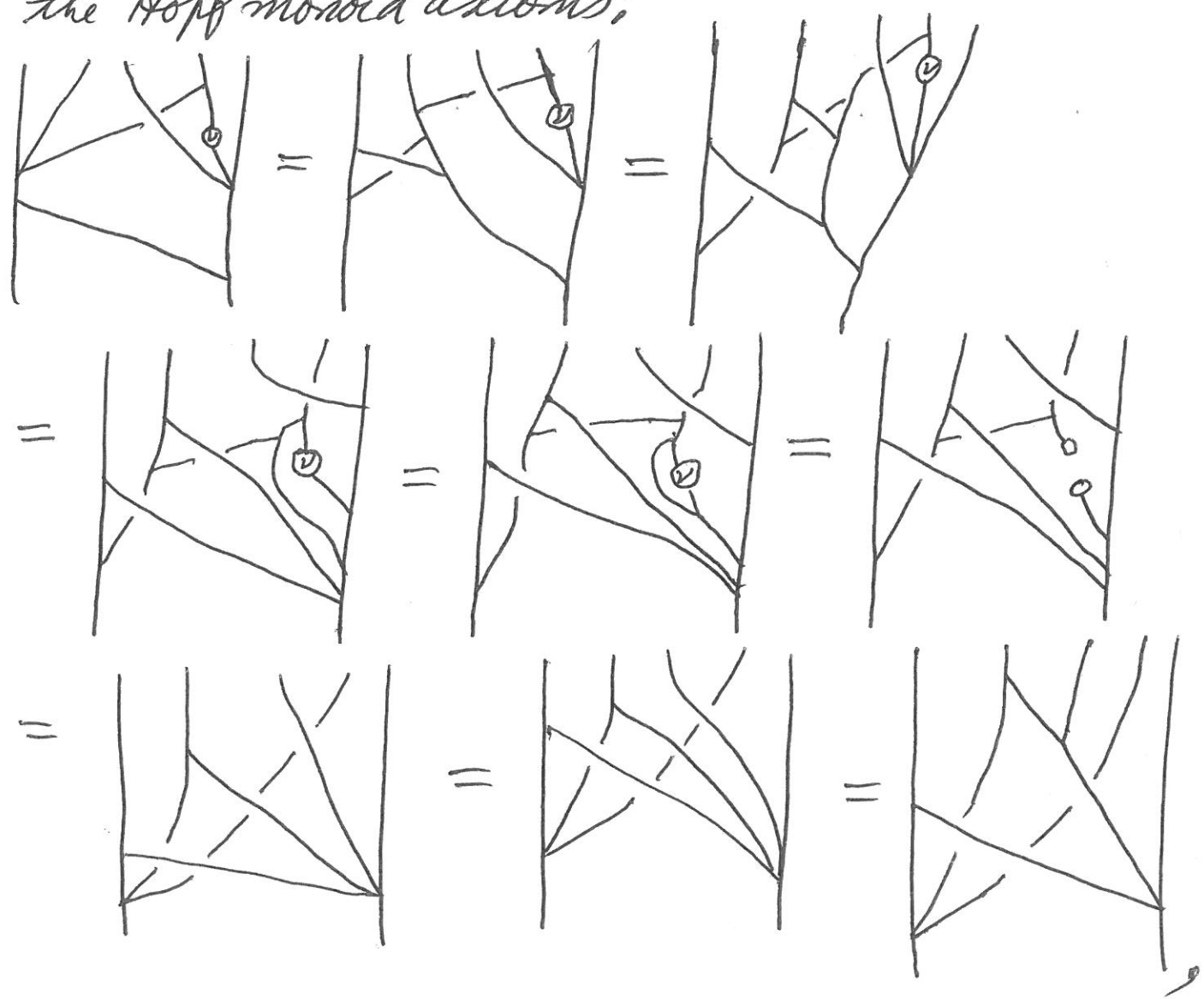
$$\begin{aligned} H^{\otimes 5} H^{\otimes 3} \otimes H &\xrightarrow{\quad} H^{\otimes 6} H^{\otimes 4} \otimes H \\ &\xrightarrow{H^{\otimes 2} \otimes c_{H, H} \otimes H^{\otimes 2}} H^{\otimes 6} \xrightarrow{\mu_3 \otimes \mu_3} H^{\otimes 2} \end{aligned}$$

The right $H \otimes H$ has action given by



$$\begin{aligned} H^{\otimes 5} \otimes H^{\otimes 4} &\xrightarrow{\quad} H^{\otimes 6} \xrightarrow{H \otimes c_{H, H} \otimes c_{H, H} \otimes H} H^{\otimes 6} \\ &\xrightarrow{H^{\otimes 2} \otimes c_{H, H} \otimes H^{\otimes 2}} H^{\otimes 6} \xrightarrow{\mu_3 \otimes \mu_3} H^{\otimes 2} \end{aligned}$$

The proof is completed by a string calculation using the Hopf monoid axioms;



as required. $\square$